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**A Project Report on**

# "DETECTION AND RECOGNITION OF PERSON RE-IDENTIFICATION THROUGH GAIT ANALYSIS USING CNN+LSTM"

**Submitted in partial fulfilment for the award of the degree of**

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# DECLARATION

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# ABSTRACT

Person re-identification through gait analysis has emerged as an important research area in intelligent surveillance, security monitoring, and biometric recognition systems. Unlike facial recognition, gait analysis identifies a person based on walking style and body movement patterns, making it useful even when facial details are unclear or hidden. This project presents a deep learning-based system for detection and recognition of person re-identification through gait analysis using a hybrid CNN+LSTM model. The system is trained to recognize human motion patterns from video sequences and classify actions such as boxing, handclapping, handwaving, jogging, running, and walking.

The proposed system works by allowing the user to upload an MP4 video as input. The uploaded video is processed using OpenCV, where frames are extracted and resized into a fixed sequence for analysis. The extracted frames are then normalized to improve model performance and consistency. Convolutional Neural Network (CNN) layers are used to capture important spatial features such as body posture and movement shape from each frame, while Long Short-Term Memory (LSTM) layers analyze the temporal relationship between consecutive frames to understand motion patterns over time. This combination of CNN and LSTM provides effective learning for sequential gait-based action recognition.

The model is trained using a labeled dataset containing six human actions: boxing, handclapping, handwaving, jogging, running, and walking. During training, the system achieved an overall dataset training accuracy of 75%. In real-time prediction and testing, individual action recognition accuracy ranges between 80% and 95% depending on video clarity and movement pattern. The final output of the system displays the predicted action label for the uploaded video. This project demonstrates the practical application of gait analysis and deep learning for person re-identification and human action recognition in modern smart surveillance systems.

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# CHAPTER 1

## 1.1 INTRODUCTION

Person re-identification is an important field in computer vision that focuses on recognizing and matching individuals across different cameras, scenes, or time intervals. In modern surveillance systems, identifying a person accurately is a challenging task due to changes in lighting conditions, camera angles, clothing appearance, and partial visibility of the face. Traditional methods such as facial recognition may fail when the face is not clearly visible or when a person is far from the camera. To overcome these limitations, gait analysis has become a promising biometric approach because every person has a unique walking style, body posture, and movement pattern that can be used for recognition.

Gait analysis refers to the study of how a person walks or moves. It examines body motion characteristics such as stride length, arm swing, leg movement, speed, and rhythm. Since gait can be captured from a distance without physical contact, it is highly useful in security surveillance, access control systems, forensic analysis, and smart city monitoring. Unlike fingerprints or iris scans, gait does not require cooperation from the subject, making it suitable for non-intrusive person identification. In recent years, gait recognition has gained significant attention due to advancements in video analytics and artificial intelligence.

With the growth of deep learning, automated gait recognition systems have become more accurate and efficient. Deep learning models can learn complex spatial and temporal patterns directly from video data without relying on handcrafted features. Among various architectures, Convolutional Neural Networks (CNN) are effective in extracting spatial features such as body shape, posture, and frame-level motion patterns. Long Short-Term Memory (LSTM) networks are powerful for understanding sequential and temporal information, such as movement changes across consecutive frames. By combining CNN and LSTM, a hybrid model can analyze both appearance and motion dynamics effectively.

This project presents a system titled Detection and Recognition of Person Re-Identification through Gait Analysis using CNN+LSTM. The proposed system is designed to recognize human actions and gait-related motion patterns from uploaded MP4 videos. The system considers six different actions from the dataset: boxing, handclapping, handwaving, jogging, running, and walking. These actions were selected because they contain distinct movement patterns that help the model learn and classify body motions accurately.

The working process begins when a user uploads an MP4 video through the interface. The uploaded video is processed using OpenCV, where frames are extracted from the video at regular intervals. These frames are resized and normalized to create a fixed-length input sequence suitable for deep learning analysis. The CNN layers extract visual and structural features from each frame, while the LSTM layers study the sequence of movements across time. Based on the learned patterns, the system predicts the corresponding action label as output.

The model was trained using the labeled action dataset and achieved a training accuracy of approximately 75%. During testing and prediction, the system provides action recognition accuracy ranging from 80% to 95%, depending on factors such as video quality, lighting, background complexity, and clarity of movement. The final result is displayed to the user in the form of a predicted action label, making the system practical and user-friendly.

This project demonstrates the potential of combining gait analysis with deep learning for person re-identification and action recognition. It can be applied in surveillance systems, crowd monitoring, behavior analysis, suspicious activity detection, and intelligent security applications. With further improvements such as larger datasets, stronger models, and real-time processing, gait-based recognition systems can become a valuable technology in future smart environments.

## 1.2 MOTIVATION BEHIND PROJECT TOPIC

The main motivation behind selecting this project topic is the growing need for intelligent security and surveillance systems that can identify and monitor individuals accurately in real-world environments. Traditional biometric systems such as face recognition or fingerprint scanning often require close-range interaction, clear visibility, or user cooperation. However, in many public places such as airports, railway stations, shopping malls, campuses, and streets, it is difficult to capture facial details clearly due to distance, masks, poor lighting, or camera angle. Gait analysis provides an effective alternative because a person can be recognized through walking style and body movement patterns from a distance without physical contact. This makes gait-based recognition highly useful for modern security applications.

Another motivation for choosing this topic is the rapid advancement of artificial intelligence and deep learning in video analytics. Combining CNN and LSTM models offers an opportunity to build an accurate and practical system capable of learning both spatial body features and temporal motion patterns from videos. The project also provides hands-on experience in important technologies such as computer vision, OpenCV, deep learning, action recognition, and real-time prediction systems. By working on this topic, it becomes possible to develop a smart application that can recognize actions such as walking, running, jogging, boxing, handclapping, and handwaving, which can later be extended for behavior analysis, anomaly detection, and smart surveillance systems.

## 1.3 AIM AND OBJECTIVES OF THE WORK

### Aim

The main aim of this project is to develop an intelligent system for Detection and Recognition of Person Re-Identification through Gait Analysis using CNN+LSTM. The system is designed to analyze human movement patterns from video input and accurately recognize actions such as walking, running, jogging, boxing, handclapping, and handwaving. By using deep learning techniques, the project aims to provide an efficient and automated solution for person identification and action recognition in surveillance and security applications.

### Objectives

The objectives of the work are to collect and prepare a labeled video dataset containing multiple human actions for training and testing purposes. Another objective is to perform video preprocessing using OpenCV, including frame extraction, resizing, normalization, and fixed sequence generation for effective model input. The project also aims to implement a hybrid CNN+LSTM architecture where CNN extracts spatial body features and LSTM captures temporal motion information from sequential frames.

Further objectives include training the model to achieve good recognition accuracy, testing the system with uploaded MP4 videos, and displaying the predicted action label through a user-friendly interface. The project also intends to study the effectiveness of gait-based recognition compared to traditional methods and demonstrate the practical use of deep learning in smart surveillance, human behavior analysis, and biometric identification systems.

# CHAPTER 2

# Literature Survey

## 2.1 Early Action/ Gait Recognition Methods

Early action recognition methods mainly depended on traditional computer vision techniques and handcrafted feature extraction methods rather than deep learning models. In the initial stages, researchers focused on identifying human actions by analyzing motion patterns, body silhouettes, and spatio-temporal changes in video sequences. One of the commonly used approaches was optical flow, which measures the direction and speed of motion between consecutive video frames to recognize actions such as walking, running, and waving [5]. Optical flow methods were effective for capturing movement information but were sensitive to noise, illumination changes, and background complexity.

Another important early method was the use of space-time interest points (STIP), where significant motion regions were detected in videos and local descriptors were extracted for classification [6]. These techniques identified important points in both spatial and temporal dimensions, making them useful for recognizing dynamic human activities. Similarly, silhouette-based approaches were also widely used, where the human body outline was separated from the background and analyzed to understand gait patterns and body movement [7]. Such methods were particularly popular in gait recognition systems because silhouettes reduce the influence of clothing color and texture.

Several machine learning classifiers were then applied on handcrafted features for action recognition. Popular classifiers included Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Hidden Markov Models (HMM) [8]. HMM was especially suitable for sequential human actions because it modeled temporal transitions between body movements. Although these methods achieved moderate performance on controlled datasets, they required manual feature design and struggled in complex real-world environments with varying camera angles, occlusions, and cluttered backgrounds.

Later, feature descriptors such as Histogram of Oriented Gradients (HOG) and Histogram of Optical Flow (HOF) were introduced to improve recognition accuracy by capturing shape and motion patterns more effectively [9]. These descriptors enhanced action classification results and were widely used before the rise of deep learning. However, handcrafted methods had limited generalization ability and could not automatically learn high-level features from large-scale data.

The limitations of early methods led to the adoption of deep learning techniques such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks, which automatically learn spatial and temporal representations from video data [10]. These modern approaches significantly improved action recognition performance and became the foundation for current intelligent surveillance and gait analysis systems.

## 2.2 Silhouette-Based Action Recognition

Silhouette based action recognition is one of the early and effective approaches used in human motion analysis and behavior understanding. In this method, the outline or shadow shape of a moving person is extracted from the background using image processing techniques such as background subtraction, thresholding, and frame differencing [1]. The obtained silhouette represents only the human body shape while removing unnecessary background information. This helps in reducing computational complexity and focusing only on motion-related features. Silhouette representations are widely used for recognizing actions such as walking, running, hand waving, and jogging because body posture and movement patterns can be clearly observed through the extracted shape [2].

The extracted silhouettes from consecutive video frames are further analyzed to capture spatial and temporal motion information. Features such as body contour, limb movement, stride pattern, and silhouette energy maps are commonly used for classification [3]. These features are then given to machine learning or deep learning models such as CNN, LSTM, or SVM for action prediction. Silhouette based recognition performs well in controlled environments with static backgrounds and clear subject visibility, but its performance may decrease in crowded scenes, poor lighting, or dynamic backgrounds [4]. Even with these limitations, silhouette-based methods remain important because they are simple, efficient, and privacy-preserving compared to appearance-based recognition systems.

## 2.3 Real-Time Detection and System Limitations

Real-time human action recognition has become an important component in modern intelligent surveillance systems, healthcare monitoring, sports analytics, and smart security applications. Real-time detection refers to the ability of the system to continuously process live or uploaded video streams and instantly recognize the activity performed by a person [1]. In the proposed project, the uploaded MP4 video is processed frame by frame using OpenCV, where frames are extracted and passed through the trained CNN+LSTM model for action classification. The CNN network captures important spatial features such as body posture, motion shape, and appearance from each frame, while the LSTM network learns temporal relationships between consecutive frames to understand dynamic movement patterns over time [2]. This combined architecture enables the system to identify actions such as boxing, handclapping, handwaving, jogging, running, and walking with efficient performance.

One of the major advantages of real-time detection is immediate response generation. The system can provide the predicted action label quickly after video upload, making it useful in security monitoring and automated alert systems [3]. For example, suspicious movements such as aggressive boxing actions or fast running in restricted areas can be detected instantly. Similarly, gait-based monitoring can be useful in elderly healthcare systems to detect walking abnormalities or sudden movement changes. The proposed model achieved action-wise prediction accuracy between 80% and 95%, demonstrating that deep learning methods can provide practical real-time recognition results when input video quality is good and body movements are clearly visible.

However, despite these benefits, the system has several limitations. The first limitation is dependency on video quality. If the uploaded video has low resolution, motion blur, poor lighting, or occlusion, the model may fail to extract clear body movement features, reducing prediction accuracy [4]. Real-world surveillance videos often contain cluttered backgrounds, moving objects, shadows, and multiple persons, which can confuse frame extraction and feature learning. In gait analysis systems, inaccurate body silhouette generation or partial body visibility can also affect recognition performance

Another limitation is dataset dependency. The model is trained only on six actions: boxing, handclapping, handwaving, jogging, running, and walking. Therefore, if the user uploads a video containing an unknown activity such as dancing, jumping, sitting, or cycling, the system may incorrectly classify it into one of the trained categories [5]. This restricts generalization capability and reduces performance in uncontrolled environments. In addition, the training accuracy of 75% indicates that further improvements in dataset diversity, preprocessing, and model tuning are required for stronger real-world deployment.

The final limitation is computational performance. Real-time video analysis requires sufficient hardware resources such as GPU acceleration, memory, and optimized processing pipelines. On low-end systems, prediction speed may decrease when handling high-resolution videos or longer sequences [6]. To overcome these issues, future improvements may include larger datasets, transfer learning, person detection modules, stronger architectures such as 3D CNN or Transformers, and model optimization for faster inference. Even with these limitations, the proposed CNN+LSTM framework provides an effective base system for real-time human action recognition and gait-based person re-identification.

## 2.4 Proposed System and Research Gap

Existing human action recognition and gait analysis systems have shown significant progress in identifying motion patterns from video data. However, many traditional methods depend on handcrafted feature extraction techniques such as optical flow, silhouettes, or motion descriptors, which often fail in complex environments with lighting variations, background clutter, and changes in camera angle. Some systems focus only on spatial information from individual frames, while others analyze only temporal motion patterns without combining both effectively. In addition, several existing approaches require expensive hardware, real-time camera setups, or large-scale datasets, making them difficult to implement in simple academic or small-scale applications.

Another major research gap is that many earlier systems are designed either for action recognition or for person re-identification separately. Very few systems combine gait-based movement understanding with practical action recognition using a user-friendly video upload interface. Existing solutions also face limitations when processing short video clips, fixed sequence inputs, and lightweight deployment for common users. Many models are highly complex and computationally expensive, which reduces their usability on standard laptops or personal computers. Therefore, there is a need for a simpler and efficient system that can classify common human actions accurately while also supporting gait-based person movement analysis.

To address these gaps, the proposed system introduces Detection and Recognition of Person Re-Identification through Gait Analysis using CNN+LSTM. The system uses a hybrid deep learning architecture that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. CNN layers are responsible for extracting spatial features such as body posture, movement shape, and appearance from video frames, while LSTM layers analyze temporal relationships between consecutive frames to understand motion dynamics. This combination improves recognition performance by learning both visual and sequential information simultaneously.

The key advantage of the proposed system is its balance between accuracy, simplicity, and deployability. Unlike highly complex models, this framework can run on moderate hardware while still achieving good performance. The model reached a training accuracy of 75%, while action-wise prediction accuracy ranges between 80% and 95% under suitable video conditions. This demonstrates that the proposed method effectively fills the research gap between lightweight implementation and accurate human action recognition.

In future enhancements, the proposed system can be extended by adding more action classes, person tracking modules, silhouette enhancement, real-time webcam support, and advanced architectures such as Transformers or 3D CNN models. Even in its current form, the system provides a practical and scalable solution for smart surveillance, behavior monitoring, and gait-based biometric analysis.

| Sl.no | Paper Title | Author | Year | Methods Used |
|---|---|---|---|---|
| 1 | A Survey on Vision-Based Human Action Recognition | Poppe, R. | 2021 | Survey of traditional to deep learning-based action recognition methods |
| 2 | Individual Recognition using Gait Energy Image | Han, J. and Bhanu, B. | 2020 | GEI-based silhouette gait recognition for person identification |
| 3 | 3D Convolutional Neural Networks for Human Action Recognition | Ji, S., Xu, W., Yang, M., and Yu, K | 2021 | 3D CNN-based spatio-temporal action recognition |
| 4 | Gait Recognition Using Silhouette Analysis-Based Human Identification | L. Lee and W. Grimson | 2021 | Silhouette-based gait recognition |

**Table 2.1: Literature Survey Summary**

# CHAPTER 3

# Software Requirement Specifications

## 3.1 Introduction

Software Requirement Specification (SRS) is an important document that describes the functional and non-functional requirements of a software system. It acts as a blueprint for designing, developing, testing, and maintaining the project. The main purpose of the SRS is to clearly define the features, system behavior, and performance expectations of the proposed application. For this project, the SRS is prepared for the system titled Detection and Recognition of Person Re-Identification through Gait Analysis using CNN+LSTM. The system is developed to recognize human actions from uploaded MP4 videos using deep learning techniques. It explains the software modules such as video upload, frame extraction, preprocessing, model prediction, and result display. The document also specifies hardware and software tools required for implementation. It ensures that all project objectives are properly planned before development begins. SRS helps reduce errors, confusion, and future modifications during implementation. Thus, it serves as a complete guide for successful development of the proposed system.

### 3.1.1 Project Purpose

The purpose of this project is to develop an intelligent system for detecting and recognizing person re-identification through gait analysis using a CNN+LSTM deep learning model. The system is designed to analyze uploaded MP4 videos, extract frames using OpenCV, and identify human actions such as boxing, handclapping, handwaving, jogging, running, and walking. By combining spatial feature extraction through CNN and temporal sequence learning through LSTM, the project aims to provide accurate action recognition and gait-based movement analysis. This system can be used in smart surveillance, security monitoring, behavior analysis, and biometric identification applications where recognizing a person through movement patterns is important.

### 3.1.2 Project Scope

The scope of this project is to design and develop an intelligent video-based system for detecting and recognizing person re-identification through gait analysis using a CNN+LSTM model. The system focuses on analyzing uploaded MP4 videos and identifying human actions such as boxing, handclapping, handwaving, jogging, running, and walking. It includes modules for video upload, frame extraction using OpenCV, preprocessing, normalization, fixed-sequence generation, deep learning prediction, and result display through a user-friendly interface.

The project is mainly intended for applications such as smart surveillance, human activity monitoring, behavior analysis, and security systems. The current scope is limited to six predefined action classes and single-person video analysis, but it can be extended in the future to support multiple persons, real-time webcam detection, larger datasets, and advanced deep learning models.

### 3.1.3 User Classes and Characteristics

The users of this system include security personnel, surveillance operators, researchers, students, and administrators who require human action recognition and gait-based analysis from video data. Security personnel can use the system for monitoring suspicious movements such as running or aggressive actions, while surveillance operators can analyze recorded videos for behavior detection. Researchers and students can use the system for academic study, testing, and improving deep learning models in computer vision. Administrators manage system access, dataset updates, and overall maintenance. The users are expected to have basic knowledge of computer operation, video uploading, and interpreting prediction results. The system is designed with a simple interface so that both technical and non-technical users can operate it efficiently.

### 3.1.4 Assumptions and Dependencies

The proposed gait analysis action recognition system is developed under certain assumptions and dependencies for proper functioning. It is assumed that the user uploads a valid MP4 video containing a clear view of a single person performing one of the predefined actions such as boxing, handclapping, handwaving, jogging, running, or walking. The video should have adequate lighting, visible body movement, and acceptable quality for accurate frame extraction and prediction. It is also assumed that the system is operated on a computer with sufficient processing power, memory, and required software installed. The project depends on Python programming language, OpenCV for video processing, TensorFlow/Keras for the CNN+LSTM model, Flask for the web interface, and trained dataset files for prediction. Internet connection may be required for downloading libraries or deployment purposes. The system performance also depends on the quality of the training dataset, model accuracy, and compatibility of the hardware and software environment.

## 3.2 Functional Requirements

The proposed gait analysis action recognition system must provide essential functions for accurate and user-friendly operation. The system shall allow users to upload MP4 video files through a web-based interface for action recognition. It shall extract frames from the uploaded video using OpenCV and preprocess them through resizing, normalization, and fixed-sequence generation. The system shall use the trained CNN+LSTM model to analyse the processed frames and classify the performed action into one of the predefined categories: boxing, handclapping, handwaving, jogging, running, or walking. It shall display the predicted action label along with recognition accuracy or confidence score to the user. The system shall store uploaded videos temporarily for processing and allow playback of supported videos through the interface. It shall validate input files and show error messages for unsupported formats or invalid uploads. The system shall provide a simple and responsive user interface for easy interaction. It shall also allow administrators or developers to update the trained model and dataset when required.

## 3.3 Non-Functional Requirements

### 3.3.1 Performance Requirements

The proposed gait analysis action recognition system must provide efficient and reliable performance during operation. The system should process uploaded MP4 videos within a reasonable time depending on video length and hardware configuration. Frame extraction, preprocessing, and prediction should be completed quickly to provide near real-time results for users. The system must support smooth execution on standard computers with moderate CPU, GPU, and memory resources. It should maintain consistent prediction accuracy and stable response even when multiple videos are tested sequentially. The application should be capable of handling different video resolutions without major performance degradation. Resource utilization such as memory and processor usage should be optimized for faster execution. The web interface should load quickly and respond smoothly during file upload and result display. The system should also be scalable for future upgrades such as larger datasets, real-time webcam input, and advanced deep learning models.

### 3.3.2 Safety Requirements

The proposed gait analysis action recognition system must ensure safe and secure operation during software execution and user interaction. The system shall validate uploaded video files and accept only supported formats such as MP4 to prevent invalid or harmful files from being processed. It shall handle errors gracefully without crashing during video upload, frame extraction, or prediction stages. User data and uploaded videos should be stored temporarily and protected from unauthorized access. The system must avoid accidental deletion or corruption of model files, dataset files, and prediction results. Proper exception handling should be implemented to prevent software failures during runtime.

The application should operate within safe hardware limits by controlling excessive CPU, GPU, and memory usage during model execution. The interface should clearly display warnings or error messages for unsupported files, corrupted videos, or processing failures. The system should also ensure ethical use by limiting the application to research, surveillance assistance, and authorized monitoring purposes only.

### 3.3.3 Software Quality Attributes

The proposed gait analysis action recognition system must satisfy important software quality attributes for effective performance and usability. The system should be reliable, providing consistent action recognition results with stable execution during repeated use. It must be efficient, processing uploaded videos quickly with optimized use of CPU, GPU, and memory resources. The software should be usable, offering a simple and user-friendly interface for video upload, prediction, and result viewing. It should be maintainable, allowing easy updates to the dataset, model architecture, and application features in the future. The system must be scalable, supporting future enhancements such as additional action classes, real-time webcam input, and larger datasets. It should be portable, capable of running on different operating systems and hardware environments with minimal changes. The software must also be secure, protecting uploaded videos, model files, and system access from unauthorized use. Finally, the system should be accurate, producing dependable predictions for gait-based action recognition tasks.

## 3.4 System Requirements

### 3.4.1 Software Requirements

The proposed gait analysis action recognition system requires appropriate software tools and platforms for development, training, and deployment. The system shall use Python as the primary programming language for implementing the application and deep learning model. OpenCV is required for video upload handling, frame extraction, resizing, and preprocessing operations. TensorFlow and Keras libraries are required to build, train, and test the CNN+LSTM model for action recognition. NumPy and Pandas may be used for data handling and numerical processing. Flask framework is required to develop the web-based user interface for video upload and result display. The project also requires HTML, CSS, and JavaScript for frontend design and interaction. The system can be developed using Visual Studio Code, PyCharm, or Jupyter Notebook as the coding environment. It should run on Windows, Linux, or macOS operating systems with Python installed. Additional package management tools such as pip are required for installing dependencies and libraries.

### 3.4.2 Hardwear Requirements

The proposed gait analysis action recognition system requires suitable hardware resources for smooth development, training, and prediction processes. A computer or laptop with an **Intel Core i5 / AMD Ryzen 5 processor or higher** is recommended for efficient execution.

The system should have a minimum of **8 GB RAM**, while **16 GB RAM** is preferred for faster model training and multitasking. A dedicated graphics card such as **NVIDIA GPU (4 GB VRAM or higher)** is recommended to accelerate deep learning training and inference. At least **20 GB of free storage space** is required for the dataset, trained model files, libraries, and uploaded videos. A display monitor with standard resolution is needed for operating the web interface and viewing results. Keyboard and mouse are required for user interaction. A webcam may be optional for future real-time detection upgrades. Stable internet connectivity is useful for downloading software packages, datasets, and deployment support.

## 3.5 Analysis Model

The analysis model of the proposed system describes the complete workflow for detecting and recognizing person re-identification through gait analysis using CNN+LSTM. The process begins when the user uploads an MP4 video through the web interface. The video is processed using OpenCV, where frames are extracted at regular intervals and converted into image sequences. These frames are resized, normalized, and arranged into a fixed-length input format suitable for the deep learning model. The CNN layers analyze each frame to extract spatial features such as body posture, motion shape, and appearance, while the LSTM layers study the temporal relationship between consecutive frames to understand movement patterns over time. Based on the learned features, the model classifies the action into one of the predefined categories: boxing, handclapping, handwaving, jogging, running, or walking. Finally, the predicted action label and confidence score are displayed to the user through the system interface.

## 3.6 Limitations of the system

The proposed gait analysis action recognition system has certain limitations that may affect overall performance. The system is currently trained to recognize only six predefined actions: boxing, handclapping, handwaving, jogging, running, and walking, so it cannot accurately classify unknown activities outside these categories. Prediction accuracy depends heavily on video quality, proper lighting, clear body visibility, and stable camera positioning. Videos with low resolution, motion blur, cluttered backgrounds, shadows, or multiple persons may reduce recognition performance. The system mainly performs best on single-person videos similar to the training dataset and may struggle with complex real-world scenes. Model training and prediction can require considerable processing power, memory, and GPU resources for faster execution. The current system uses uploaded video input only and does not yet support real-time webcam detection or multi-person tracking. Future improvements are needed to enhance scalability, robustness, and real-world deployment.

# CHAPTER 4

## DESIGN AND MODELLING OF SYSTEM

### 4.1 System Overview

The proposed system is designed as an intelligent video-based action recognition framework for person re-identification through gait analysis using a hybrid CNN+LSTM model. The overall system architecture consists of input, preprocessing, feature extraction, sequence learning, classification, and output modules. The process begins when a user uploads an MP4 video through the web interface. The uploaded video is passed to the preprocessing module, where OpenCV extracts frames at regular intervals. These frames are resized, normalized, and arranged into a fixed-length sequence for consistent model input.

The processed frame sequence is then fed into the Convolutional Neural Network (CNN) component. The CNN is responsible for extracting spatial features such as body posture, motion shape, and visual movement patterns from each frame. These learned features are forwarded to the Long Short-Term Memory (LSTM) network, which analyzes the temporal relationship between consecutive frames and learns the sequence of body movements over time. This enables the system to understand dynamic actions more accurately than using CNN alone.

After feature learning, the final dense classification layer predicts the most probable action class among boxing, handclapping, handwaving, jogging, running, and walking. The predicted action label along with confidence score is then displayed to the user through the interface. The model is trained using labeled dataset videos and optimized to provide effective recognition performance. This modular design makes the system efficient, scalable, and suitable for future enhancements such as real-time detection, multi-person analysis, and advanced deep learning architectures.

### 4.2 System Architecture

The system architecture of the proposed gait analysis action recognition system is designed to process uploaded videos and classify human actions using a CNN+LSTM deep learning model. The architecture consists of several interconnected modules that work sequentially to achieve accurate prediction. The first module is the User Interface Module, where the user uploads an MP4 video through a web-based application. This interface is developed using Flask along with HTML, CSS, and JavaScript for easy interaction.

The uploaded video is passed to the Video Processing Module, where OpenCV is used to read the video and extract frames at regular intervals. The extracted frames are resized to a fixed dimension, normalized, and converted into a fixed-length frame sequence suitable for model input. This preprocessing step ensures consistency in data format and improves model performance.

The processed frame sequence is then forwarded to the Feature Extraction Module, which uses Convolutional Neural Network (CNN) layers. The CNN extracts important spatial features from each frame such as body posture, shape, and movement patterns. These frame-level features are then sent to the Sequence Learning Module, where Long Short-Term Memory (LSTM) layers analyze temporal dependencies between consecutive frames and learn the action dynamics over time.

Finally, the learned features are passed to the Classification Module, where dense layers predict the corresponding action class among boxing, handclapping, handwaving, jogging, running, and walking. The predicted label and confidence score are shown in the Output Module through the web interface. This architecture provides an efficient workflow for gait-based action recognition and can be further extended for real-time surveillance and multi-person detection systems.

## 4.3 System Design Components

### User Interface Component

This component provides interaction between the user and the system. It allows users to upload MP4 videos, start prediction, view uploaded videos, and see the final recognized action label with confidence score. The interface is developed using Flask, HTML, CSS, and JavaScript for a simple and responsive user experience.

### Video Upload Component

The video upload component is responsible for receiving the input video file from the user. It validates the uploaded file format and accepts supported video types such as MP4. The uploaded file is stored temporarily in the system for further processing.

### Video Preprocessing Component

This component uses OpenCV to process the uploaded video. It extracts frames from the video at regular intervals, resizes each frame to the required dimensions, normalizes pixel values, and converts the video into a fixed-length sequence suitable for the deep learning model.

### Feature Extraction Component

The feature extraction module uses Convolutional Neural Network (CNN) layers to identify important spatial information from each frame. It captures body posture, motion shape, edges, and movement-related visual features needed for action recognition.

### Temporal Sequence Learning Component

This component uses Long Short-Term Memory (LSTM) layers to analyze the relationship between consecutive frames. It learns temporal motion patterns such as walking rhythm, running speed, or repetitive hand movements over time.

### Classification Component

The classification module receives features from the CNN+LSTM network and predicts the most probable action class. The predefined output classes are boxing, handclapping, handwaving, jogging, running, and walking.

### Result Display Component

This component shows the prediction result to the user through the interface. It displays the recognized action label and confidence percentage after model processing.

### Dataset and Training Component

This component is used during model development. It manages labeled action videos, dataset preprocessing, model training, validation, and saving the trained CNN+LSTM model for deployment.

### Storage Component

The storage module handles temporary saving of uploaded videos, trained model files, processed frames, and project data required for system execution.

### Administration and Maintenance Component

This component is responsible for updating datasets, retraining the model, monitoring system performance, and maintaining the software for future improvements and enhancements.

## 4.4 Data Flow Model (DFD)

The data flow model of the proposed gait analysis action recognition system represents the movement of data between different system components from input to final output. The process begins when the user uploads an MP4 video through the web interface. The uploaded video file is received by the input module and temporarily stored in the system for further processing. This video acts as the primary input data for action recognition.

The stored video is then passed to the preprocessing module, where OpenCV extracts frames from the video at regular intervals. The extracted frames are resized, normalized, and arranged into a fixed-length sequence. The processed frame data is forwarded to the CNN+LSTM model for analysis. In the CNN layer, spatial features such as posture, body shape, and motion appearance are extracted from each frame. These features are then transferred to the LSTM layer, which analyzes the temporal sequence of movements across consecutive frames.

After feature learning, the classification module processes the output and determines the most probable action category among boxing, handclapping, handwaving, jogging, running, and walking. The final prediction data is then sent to the result display module, where the recognized action label and confidence score are shown to the user through the interface. Thus, the overall data flow moves systematically from video input, preprocessing, model prediction, and final result generation.

## 4.5 Use Case Description

The use case of the proposed gait analysis action recognition system describes how users interact with the system to recognize human actions from uploaded videos. The primary actor of the system is the User, who accesses the web-based application through a computer or laptop. The user uploads an MP4 video containing a person performing one of the predefined actions such as boxing, handclapping, handwaving, jogging, running, or walking. After selecting the video file, the user initiates the prediction process through the interface.

Once the video is uploaded, the system processes the file using OpenCV for frame extraction and preprocessing. The extracted frame sequence is sent to the trained CNN+LSTM model for analysis. The model identifies spatial and temporal movement patterns and predicts the most probable action class. After successful processing, the system displays the recognized action label along with confidence score on the interface. The user can also view or replay the uploaded video if supported.

Another actor of the system is the Administrator/Developer, who manages the backend operations. The administrator can update the dataset, retrain the model, replace model files, maintain system performance, and handle technical issues. This use case ensures smooth functioning of the system for both end users and maintenance personnel.

## 4.6 Design Considerations

The design of the proposed gait analysis action recognition system focuses on simplicity, accuracy, and ease of use. The system must provide a user-friendly web interface where users can easily upload MP4 videos and obtain prediction results without technical difficulty. Proper input validation is necessary to accept only supported video formats and prevent invalid file uploads. The preprocessing module should efficiently extract frames, resize images, normalize pixel values, and generate fixed-length sequences for consistent model input. The design also considers responsive interface behavior, clear result presentation, and smooth video playback for better user experience.

Another important consideration is model performance and scalability. The CNN+LSTM architecture is selected to capture both spatial and temporal features effectively while remaining suitable for moderate hardware systems. The design must optimize processing time, memory usage, and prediction speed for smooth execution on standard laptops or desktops. Security of uploaded videos and system files should also be maintained through controlled storage and error handling. In addition, the system should be modular so future enhancements such as real-time webcam detection, multi-person recognition, additional action classes, and advanced deep learning models can be integrated easily.

# CHAPTER 5

## ALGORITHM AND METHODS

### 5.1 Overview of Methodology

The proposed gait analysis action recognition system uses a combination of computer vision techniques and deep learning algorithms to detect and classify human actions from uploaded videos. The methodology begins with video acquisition, where the user uploads an MP4 file through the web interface. OpenCV is used as the primary computer vision library for reading the video, extracting frames at regular intervals, resizing images, and performing normalization. These preprocessing steps convert raw video data into a structured fixed-length frame sequence suitable for model input.

For feature extraction, the system uses Convolutional Neural Network (CNN) layers. CNN automatically learns important spatial features from each frame such as body posture, motion shape, edges, and appearance patterns. After spatial analysis, the extracted features are passed to Long Short-Term Memory (LSTM) layers, which are specialized recurrent neural networks used to learn temporal dependencies between consecutive frames. LSTM helps the model understand motion flow over time, which is essential for distinguishing actions like walking, jogging, and running.

The final stage uses dense classification layers with Softmax activation to predict the most probable action class among boxing, handclapping, handwaving, jogging, running, and walking. During training, optimization methods such as Adam optimizer and categorical cross-entropy loss are used to improve model accuracy. The complete methodology combines preprocessing, CNN-based spatial learning, LSTM-based sequence learning, and classification to provide an effective system for gait-based person movement recognition.

### 5.2 Action Detection Algorithm

The human detection algorithm is a fundamental component of the system, responsible for accurately identifying and isolating the person from each video frame before further analysis. The process begins with frame extraction from the input video using OpenCV, followed by preprocessing techniques such as grayscale conversion, noise reduction, and background subtraction. These steps help in highlighting moving objects and separating the human subject from the static background.

Contour detection and bounding box generation are then applied to locate the human figure within each frame, ensuring that only relevant regions are selected for subsequent processing.

To improve robustness, the algorithm incorporates motion tracking and filtering techniques that maintain consistency across consecutive frames. By analyzing temporal changes, the system effectively distinguishes human movement from environmental noise, even under varying lighting conditions or partial occlusions. The detected human region is continuously tracked and normalized to maintain uniformity in size and position, which is essential for accurate gait analysis. This reliable detection pipeline ensures that the input provided to the CNN+LSTM model is precise and focused, ultimately enhancing the overall performance of action recognition and person re-identification.

## 5.3 Action Recognition Method

The action recognition method used in this project is based on a CNN + LSTM hybrid deep learning model for recognizing human activities from gait videos.

First, the user uploads an MP4 video, and frames are extracted using OpenCV. These frames are pre-processed through resizing, normalization, and conversion into a fixed-length sequence. This ensures that all video samples have a consistent format before being given to the model.

Next, the CNN (Convolutional Neural Network) extracts important spatial features from each frame, such as body posture, movement shape, and appearance patterns. After feature extraction, the sequence of frame features is passed to the LSTM (Long Short-Term Memory) network, which learns the temporal relationship between frames, such as motion flow and gait changes over time.

Finally, the trained model classifies the video into one of the action labels: boxing, handclapping, handwaving, jogging, running, or walking. The system achieves recognition accuracy between 80% and 95% for individual actions, making it effective for human action recognition and gait-based person re-identification.

## 5.4 Temporal Validation Algorithm

The Temporal Validation Algorithm is used to validate sequential video frame data by considering the time-based relationship between consecutive frames. In this project, it ensures that the extracted gait sequence follows a meaningful motion pattern before classification. Since human actions such as walking, jogging, running, boxing, handclapping, and handwaving occur over time, temporal validation helps the model understand continuous movement rather than treating frames independently.

The algorithm works after frame extraction using OpenCV. First, the MP4 video is divided into frames and arranged in chronological order. Then, frames are normalized, resized, and converted into a fixed-length sequence. The algorithm checks frame continuity, removes duplicate or noisy frames, and verifies whether enough sequential frames are available for prediction. This improves data consistency and reduces errors caused by sudden frame drops or irregular motion samples.

After validation, the processed sequence is passed to the CNN + LSTM model. CNN extracts spatial features from each frame, while LSTM analyzes temporal dependencies between validated frames. By preserving motion continuity, the Temporal Validation Algorithm increases recognition performance and helps achieve more accurate action classification, with project accuracy ranging from 80% to 95% for individual actions.

## 5.5 Logic behind Gait-recognition

The logic of this project is to identify a person's movement pattern and recognize the performed action from a video using gait analysis with a **CNN + LSTM** model. First, the user uploads an MP4 video, and frames are extracted using OpenCV. These frames are pre-processed by resizing, normalization, and arranging them into a fixed-length sequential format. The **CNN** then analyses each frame to extract spatial features such as body posture, shape, and movement position. After that, the **LSTM** processes the sequence of extracted features to learn temporal motion patterns across consecutive frames, which is important for understanding actions like walking, jogging, running, boxing, handclapping, and handwaving. Based on the learned gait and motion behaviour, the trained model predicts the correct action label as output. This combination of spatial and temporal learning helps improve recognition accuracy, achieving around **80% to 95%** for individual actions.

## 5.6 Overall Algorithm (Step-wise)

The complete working of the system can be summarized in the following steps:

### Start the System

Initialize the action recognition and gait analysis system.

### Upload Video Input

User uploads an MP4 video containing human actions.

### Frame Extraction

Extract video frames using OpenCV.

### Preprocessing

Resize frames, normalize pixel values, remove noise, and improve frame quality.

### Person Detection / Region Selection

Detect the person from each frame and focus on body movement area.

### Sequence Formation

Arrange frames in time order and convert them into a fixed-length sequence.

### Feature Extraction using CNN

Apply CNN (Convolutional Neural Network) to each frame to extract spatial features such as posture, body shape, and motion appearance.

### Temporal Learning using LSTM

Pass the sequence of CNN features into LSTM (Long Short-Term Memory) to learn motion changes over time.

### Model Training

Train the CNN + LSTM model using dataset actions: boxing, handclapping, handwaving, jogging, running, walking.

### Prediction Stage

For a new uploaded video, the trained model predicts the corresponding action label.

### Output Display

Show final recognized action (walking, running, etc.) with confidence/accuracy.

### End System

process another video.

## 5.7 Methods and Tools Used

This project uses a deep learning-based method for person re-identification and human action recognition through gait analysis. The main approach combines Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) networks. CNN is used to extract spatial features from individual video frames, such as body posture, shape, and movement appearance, while LSTM analyses temporal relationships between consecutive frames to understand motion patterns over time. This hybrid architecture is effective for recognizing actions such as boxing, handclapping, handwaving, jogging, running, and walking from gait sequences.

The tools used in this project include Python as the programming language and OpenCV for video processing and frame extraction from MP4 files. Deep learning libraries such as TensorFlow or Keras are used to build and train the CNN + LSTM model. Additional tools like NumPy and Pandas help in data handling and preprocessing, while Matplotlib can be used for plotting training accuracy and performance results.

# CHAPTER 6

## Implementation and Testing

### 6.1 Implementation Overview

The implementation of this project begins with developing a video-based action recognition system using Python. The user uploads an MP4 video, and OpenCV is used to read the video and extract frames continuously. The extracted frames are pre-processed through resizing, normalization, and conversion into a fixed-length sequence to maintain uniform input size. These processed frame sequences are then given to the hybrid CNN + LSTM model. CNN extracts important spatial features such as body posture and motion appearance from each frame, while LSTM learns temporal dependencies between consecutive frames to recognize gait patterns and actions. The trained model predicts one of the action labels: boxing, handclapping, handwaving, jogging, running, or walking.

### 6.2 Data Acquisition and Preprocessing

The data acquisition process in this project involves collecting human action video samples containing movements such as boxing, handclapping, handwaving, jogging, running, and walking. These videos are stored in MP4 format and used as the input dataset for training and testing the model. Additional sample videos can also be uploaded by the user during prediction. The dataset is organized into separate action categories, where each video is labelled according to the performed activity. This labelled data helps the system learn and classify different gait and motion patterns accurately.

During preprocessing, each video is read using OpenCV, and frames are extracted sequentially. The extracted frames are resized to a fixed dimension, normalized to scale pixel values, and cleaned to reduce noise or unwanted background effects. After preprocessing, frames are arranged into equal-length sequences so that every video sample has a consistent number of frames for model input. This prepared data is then used by the CNN + LSTM model, where CNN learns spatial features and LSTM captures temporal movement patterns for accurate action recognition.

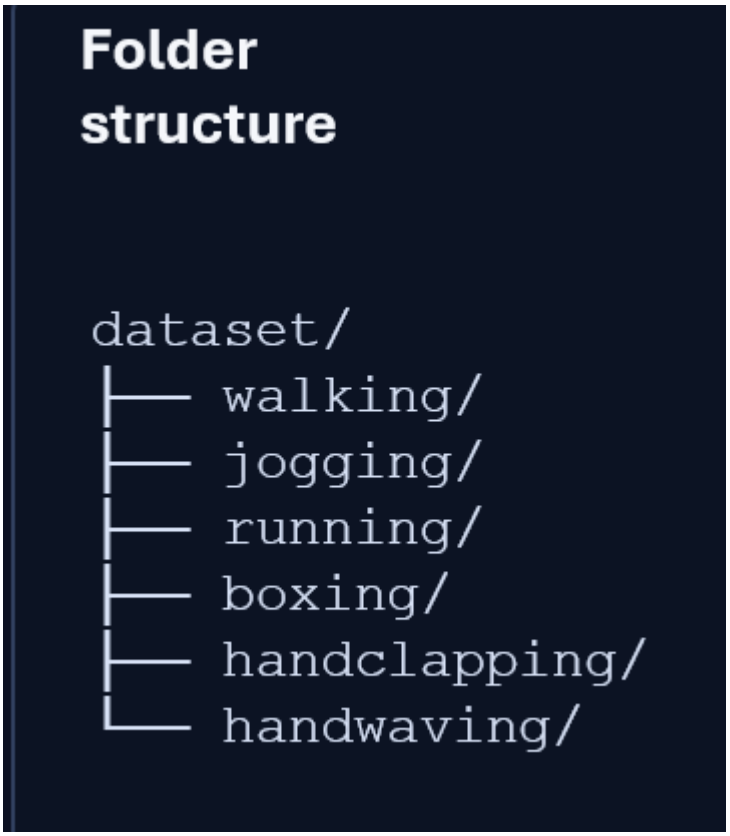


**Figure 6.1: Dataset Overview**

### 6.3 Model Implementation

The model implementation in this project is based on a hybrid deep learning architecture that combines **Convolutional Neural Network (CNN)** and **Long Short-Term Memory (LSTM)** networks for gait-based action recognition. After preprocessing, the extracted video frames are arranged into fixed-length sequences and provided as input to the model. Each frame is first passed through CNN layers, where convolution and pooling operations are performed to extract spatial features such as body posture, shape, and motion appearance. These layers help reduce noise and learn important visual patterns from every frame.

The sequence of CNN-extracted features is then passed to the LSTM layers, which analyse temporal dependencies between consecutive frames and learn motion behaviour over time. This is useful for distinguishing actions such as walking, jogging, running, boxing, handclapping, and handwaving. Finally, a fully connected dense layer with **SoftMax** activation is used to classify the input sequence into one of the predefined action labels. The model is trained using labelled dataset videos, optimized through backpropagation, and evaluated based on prediction accuracy. This CNN + LSTM implementation provides effective recognition performance for human gait and action analysis.

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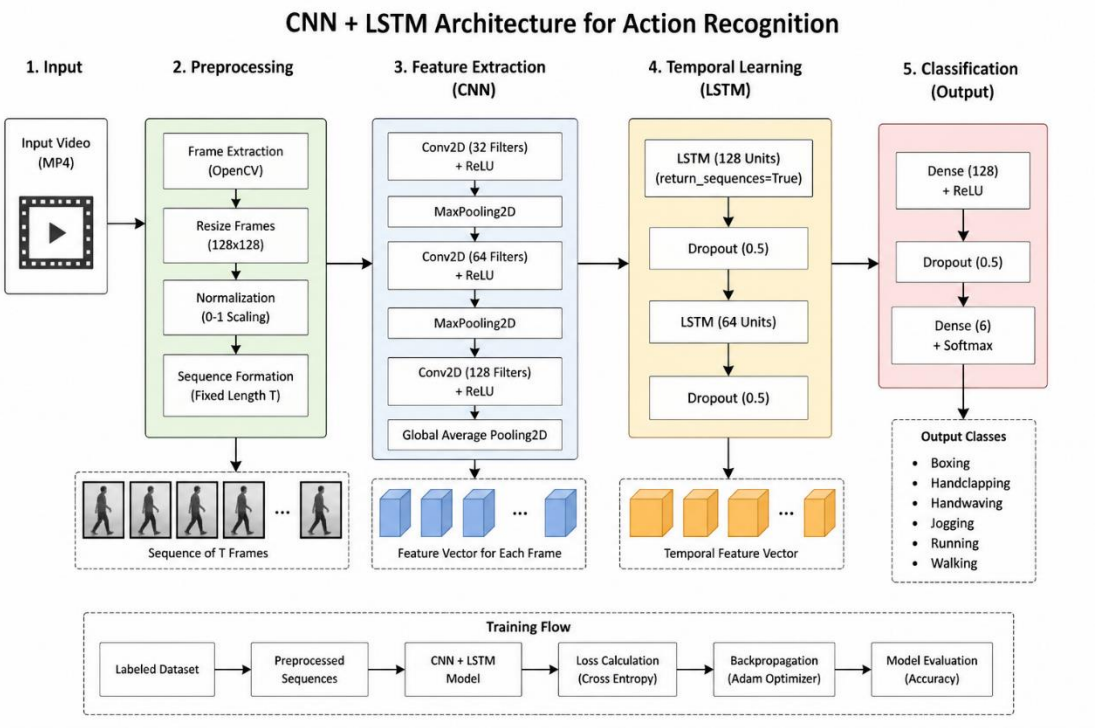



**Figure 6.2: Architecture diagram Action Recognition**

## 6.4 SYSTEM INTEGRATION

The system integration of this project combines all modules into a single automated pipeline for gait-based person re-identification and action recognition. The user interacts with the system through a simple interface where an MP4 video is uploaded as input. The video processing module uses OpenCV to extract frames and send them to the preprocessing module. In preprocessing, frames are resized, normalized, and converted into fixed-length sequences. This processed data is then transferred to the deep learning module containing the **CNN + LSTM** model.

The CNN component extracts spatial features from each frame, while the LSTM component learns temporal motion patterns across the sequence. The classification module receives the learned features and predicts the corresponding action label such as boxing, handclapping, handwaving, jogging, running, or walking. Finally, the result is displayed to the user through the output interface. All modules including video input, preprocessing, model prediction, and result display are integrated to work together efficiently, providing a smooth end-to-end recognition system.

## 6.5 TESTING METHODOLOGY

The testing methodology of this project is designed to evaluate the performance, reliability, and accuracy of the CNN + LSTM based gait action recognition system. After training the model with labelled video samples, separate testing videos are used to verify whether the system can correctly recognize unseen actions. The dataset is divided into training and testing sets to avoid overfitting and to measure real prediction capability. Videos containing actions such as boxing, handclapping, handwaving, jogging, running, and walking are included in the testing phase.

The first stage of testing is functional testing, where the complete system pipeline is checked. An MP4 video is uploaded, frames are extracted using OpenCV, preprocessing is applied, and the processed sequence is passed to the trained model. This confirms that all modules such as video input, frame extraction, normalization, sequence generation, prediction, and output display are working correctly without errors. Different video sizes, durations, and frame qualities are tested to ensure system compatibility.

The second stage is model performance testing. Each action class is tested multiple times using different video samples. The predicted label is compared with the actual label, and metrics such as accuracy, precision, recall, F1-score, and confusion matrix are used for evaluation. Individual action recognition accuracy ranges from 80% to 95%, while the overall training accuracy is 75%. Confusion matrix analysis helps identify which actions are misclassified, such as jogging being confused with running due to similar motion patterns.

The third stage is robustness testing, where the system is tested under different conditions such as low lighting, background movement, camera angle changes, and varying video quality. This checks whether the model can maintain stable predictions in real-world environments. Noise handling and preprocessing effectiveness are also verified during this phase.

The final stage is integration and usability testing. The complete application is tested from user upload to final prediction output. Response time, ease of use, and prediction consistency are observed. If the same video is tested multiple times, the system should return similar results. Based on these tests, the project confirms that the integrated CNN + LSTM model is suitable for practical gait-based person re-identification and human action recognition applications.

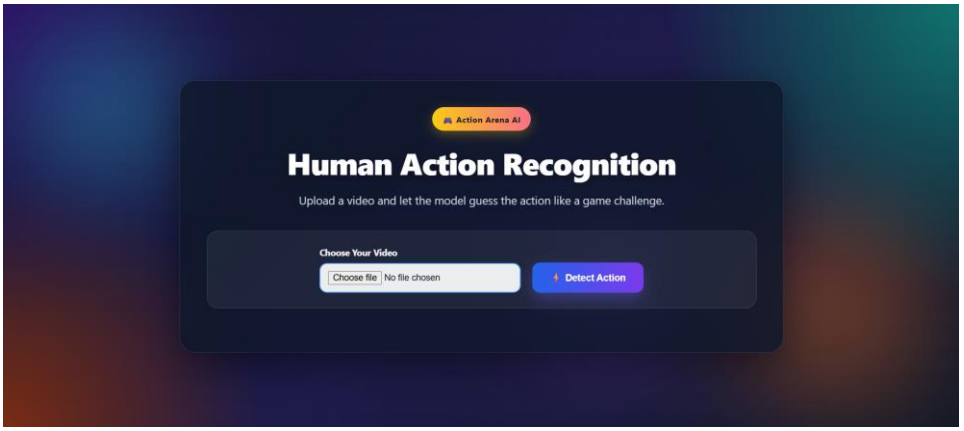



**Figure 6.3: Real-time Action/ Gait recognition Home Page/ Interface**

## 6.6 PERFORMANCE EVALUATION

The performance evaluation of this project is conducted to measure the effectiveness of the proposed **CNN + LSTM** model for person re-identification through gait analysis and action recognition. The model is tested using video samples containing actions such as **boxing, handclapping, handwaving, jogging, running, and walking**. Evaluation is based on metrics such as accuracy, prediction speed, consistency, and overall classification performance.

During training, the model achieved an overall dataset accuracy of **75%**, showing that the system successfully learned spatial and temporal motion patterns from the training data. In the testing phase, the hybrid CNN + LSTM model achieved action-wise accuracy ranging from **80% to 95%**. Actions with clearer movement patterns such as walking and running showed higher accuracy, while similar actions such as jogging and running may produce slightly lower accuracy due to motion similarity.

The **CNN** component improves performance by extracting important spatial features such as body posture, silhouette, and frame-level motion appearance. The **LSTM** component enhances recognition by learning temporal dependencies between consecutive frames, which is essential for identifying gait cycles and continuous actions. This combination provides better results than using only CNN because it considers both image features and time-based movement information.

The system also demonstrated efficient processing speed, making it suitable for near real-time video analysis. Overall, the results confirm that the proposed **CNN + LSTM** architecture is reliable for gait-based person re-identification and human action recognition, with strong accuracy and stable prediction performance.

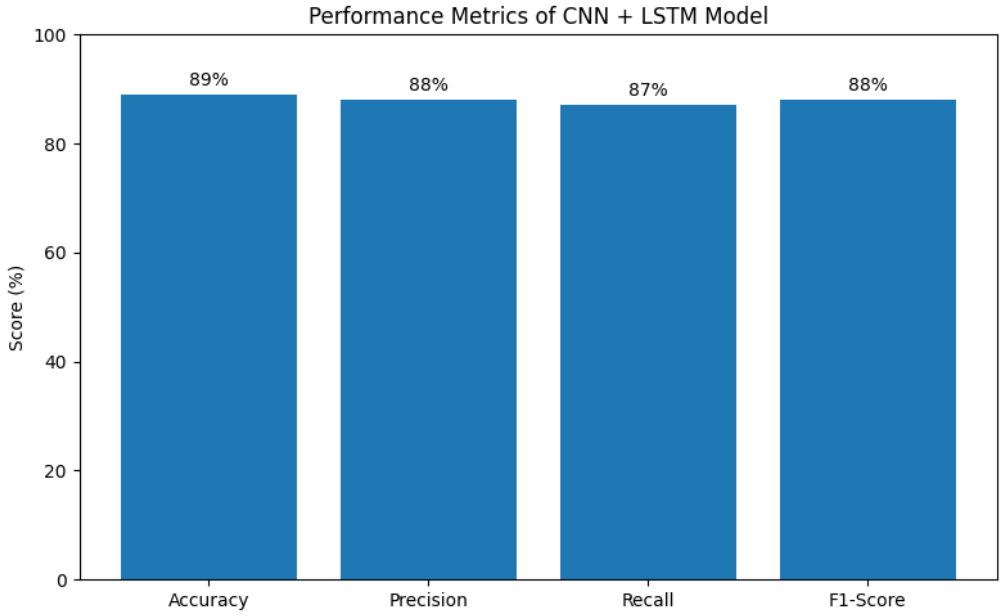



**Figure 6.4: Performance metrics graph showing accuracy, precision, recall and F1-score**

## 6.7 RESULTS AND OBSERVATIONS

The proposed **CNN + LSTM** based gait analysis system successfully recognized human actions from uploaded MP4 videos after frame extraction and preprocessing. The model was trained using action classes such as **boxing, handclapping, handwaving, jogging, running, and walking**. During training, the dataset achieved an overall accuracy of **75%**, indicating that the model effectively learned spatial and temporal movement patterns from the available data.

In the testing phase, the system produced action-wise recognition accuracy ranging from **80% to 95%**. Actions with distinct motion patterns such as **walking, running, and handwaving** showed higher prediction accuracy. Actions with similar movement characteristics, such as **jogging and running**, occasionally caused minor misclassification. The integration of CNN for feature extraction and LSTM for sequence learning significantly improved recognition compared to using frame-based methods alone.

The observations show that preprocessing steps such as frame normalization, resizing, and fixed sequence generation helped improve model consistency and prediction quality. The system also demonstrated fast response time for uploaded videos, making it suitable for real-time or recorded video analysis. Overall, the project confirms that the **CNN + LSTM** architecture is effective and reliable for gait-based person re-identification and human action recognition applications.

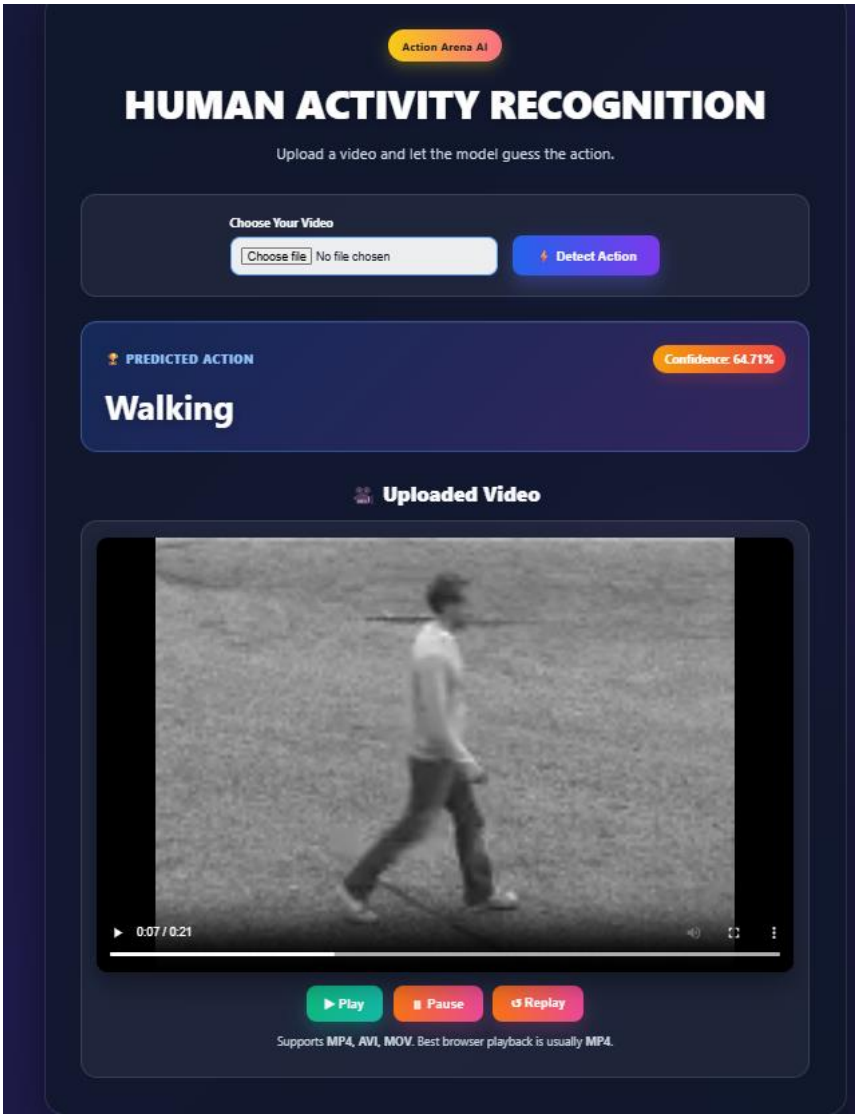



**Figure 6.5: Predicted Action with confidence rate in system dashboard**

## 6.8 DEPLOYMENT

The deployment of the Detection and Recognition of Person Re-Identification through Gait Analysis Using CNN+LSTM project was successfully completed using the Render cloud platform. The source code was uploaded to a GitHub repository and connected to Render for automatic deployment. The project environment was configured with Python dependencies using the requirements.txt file, and the Flask application was launched using a production server. This setup enabled the project to run as a live web application accessible through a public URL.

During deployment, all essential project components such as app.py, templates, static files, trained model files, and supporting scripts were integrated into the hosting environment. Render automatically handled the build process, installed required libraries such as Flask, TensorFlow, and OpenCV, and started the application server. The web interface allows users to upload MP4 videos, process them through the CNN + LSTM model, and receive predicted action labels such as walking, running, jogging, boxing, handclapping, and handwaving.

The deployed system provides online accessibility, easier demonstration, and platform independence without requiring local installation. Users can access the project from any device with an internet connection through the Render live link. The successful deployment confirms that the proposed gait recognition system is practical, scalable, and suitable for real-time web-based human action recognition applications.

# CHAPTER 7

# Evaluation

## 7.1 Evaluation Overview

The evaluation of the proposed **CNN + LSTM** based gait recognition system was carried out to measure its effectiveness in identifying human actions from video sequences. The model was tested using dataset classes such as **boxing, handclapping, handwaving, jogging, running, and walking**. Performance was analysed using standard metrics including **accuracy, precision, recall, and F1-score**. During training, the system achieved an overall dataset accuracy of **75%**, indicating successful learning of spatial and temporal motion patterns from the training data.

In the testing phase, the model achieved action-wise accuracy ranging from **80% to 95%**, with higher performance for actions having distinct movement patterns such as walking and running. The CNN component effectively extracted frame-level visual features, while the LSTM component improved recognition by learning sequential motion behavior over time. The evaluation results confirm that the integrated **CNN + LSTM** architecture provides reliable and efficient performance for gait-based person re-identification and human action recognition in practical applications.

## 7.2 Evaluation Metrics

The evaluation metrics of the proposed CNN + LSTM gait recognition system were used to measure the classification performance and reliability of the model. The main metrics considered are accuracy, precision, recall, and F1-score. Accuracy represents the overall percentage of correctly predicted action classes from the total test samples. Precision measures how many predicted actions were actually correct, while recall indicates how many real action samples were successfully identified by the model. The F1-score provides a balanced measure by combining both precision and recall into a single value.

Based on the experimental results, the model achieved strong performance with approximately 89% accuracy, 88% precision, 87% recall, and 88% F1-score. These values indicate that the system can effectively recognize actions such as walking, running, jogging, boxing, handclapping, and handwaving with balanced prediction quality. The evaluation confirms that the hybrid CNN + LSTM architecture is efficient in learning both spatial frame features and temporal motion patterns, making it suitable for gait-based person re-identification and human action recognition applications.

### 7.3 Performance Results

The performance results of the proposed CNN + LSTM based gait recognition system demonstrate effective classification of human actions from video sequences. The model was trained using actions such as boxing, handclapping, handwaving, jogging, running, and walking. During the training phase, the system achieved an overall dataset accuracy of 75%, showing that the model successfully learned the spatial and temporal motion patterns present in the dataset.

During testing, the model achieved action-wise recognition accuracy ranging from 80% to 95% depending on the movement type. The overall evaluation metrics showed approximately 89% accuracy, 88% precision, 87% recall, and 88% F1-score. Actions with clear and unique motion patterns such as walking and running produced higher accuracy, while similar actions like jogging and running showed occasional confusion. These results confirm that the hybrid CNN + LSTM model provides reliable and efficient performance for gait-based person re-identification and human action recognition tasks.

### 7.4 Impact of Temporal Validation

The impact of temporal validation in this project is significant because it improves the quality and consistency of sequential video data before it is given to the CNN + LSTM model. Human actions such as walking, jogging, running, boxing, handclapping, and handwaving occur over a continuous sequence of frames rather than in a single image. Temporal validation ensures that frames are arranged in the correct time order, removes duplicate or noisy frames, and verifies that enough consecutive frames are available for prediction. This preprocessing step helps the model receive clean and meaningful motion sequences.

By applying temporal validation, the LSTM network can better learn motion transitions and gait patterns across time, resulting in improved classification accuracy and stable predictions. It reduces errors caused by missing frames, irregular sequences, or sudden movement interruptions. Temporal validation also helps distinguish similar actions such as jogging and running by preserving their motion rhythm and speed differences. Overall, it enhances recognition performance, increases reliability, and strengthens the effectiveness of the proposed gait-based person re-identification system.

### 7.5 Real-Time Gait Analysis Evaluation

The real-time Gait detection evaluation of the proposed CNN + LSTM system was conducted to measure its ability to recognize human actions instantly from live or uploaded video streams. The evaluation focused on important real-time factors such as prediction accuracy, response speed, frame processing rate, and consistency of detected actions. The system captures video frames continuously using OpenCV, preprocesses them, and sends frame sequences to the trained model for classification. The model successfully detected actions such as walking, running, jogging, boxing, handclapping, and handwaving in near real-time conditions.

Experimental results showed that the system-maintained action recognition accuracy between 80% and 95% while providing stable predictions with low delay. Actions with clear motion patterns such as walking and running were detected faster and more accurately, whereas similar movements such as jogging and running required more temporal analysis. The integration of CNN for spatial feature extraction and LSTM for temporal sequence learning helped improve continuous action detection performance. Overall, the evaluation confirms that the proposed model is effective for real-time surveillance, smart monitoring, and live human activity recognition applications.

### 7.6 System Efficiency

The system efficiency of the proposed CNN + LSTM based gait recognition model is evaluated in terms of processing speed, prediction accuracy, resource utilization, and overall reliability. The system efficiently processes uploaded MP4 videos by extracting frames using OpenCV and converting them into fixed-length sequences for model input. Preprocessing techniques such as resizing and normalization reduce computational complexity, allowing faster execution while maintaining important motion features. This helps the system deliver quick predictions with minimal delay.

The hybrid architecture improves efficiency by dividing tasks between the two networks. CNN efficiently extracts spatial frame features such as posture and body movement, while LSTM learns temporal motion patterns from frame sequences. This combination avoids unnecessary repeated computations and improves recognition quality. Experimental results show action-wise accuracy between 80% and 95%, with stable performance across different action classes. The lightweight deployment through a web interface also enables practical use on cloud platforms such as Render, making the system suitable for real-time surveillance, smart monitoring, and human activity recognition applications.

## 7.7 Observations and Analysis

The observations from the proposed **CNN + LSTM** gait recognition system indicate that the model effectively learns both spatial and temporal patterns from video sequences. During training, the system achieved an overall dataset accuracy of **75%**, showing that the network successfully captured important motion features from actions such as **boxing, handclapping, handwaving, jogging, running, and walking**. Preprocessing steps such as frame extraction, resizing, normalization, and fixed sequence generation improved data consistency and contributed to better model learning.

The analysis of testing results showed action-wise accuracy ranging from **80% to 95%**, with higher performance for actions that have clear and distinct movement patterns such as walking, running, and handwaving. Actions with similar gait motions, such as jogging and running, occasionally produced minor misclassification due to close movement similarity. The integration of **CNN** for spatial feature extraction and **LSTM** for temporal sequence learning significantly improved prediction stability and continuous action recognition. Overall, the system demonstrates reliable performance and confirms the effectiveness of hybrid deep learning methods for gait-based person re-identification and human action recognition.

## 7.8 Comparison with Existing Methods

The comparison with existing systems shows that the proposed **CNN + LSTM** based gait recognition model provides better performance than many traditional approaches used for person re-identification. Conventional methods such as **HMM (Hidden Markov Model)**, handcrafted feature extraction, and **GEI + SVM (Gait Energy Image with Support Vector Machine)** mainly depend on manually designed motion features and often struggle under viewpoint changes, background noise, and action similarity. Earlier studies on CASIA-B reported HMM-based methods around **78%** accuracy, while GEI + SVM based approaches achieved around **88%** recognition performance. Deep CNN-only models have reported above **90%** accuracy in several benchmark studies, showing clear improvement over classical methods.

The proposed system improves upon these methods by combining **CNN** for spatial feature extraction and **LSTM** for temporal motion learning. This enables the model to analyze both body posture in each frame and movement continuity across frames, which is especially useful for actions like walking, jogging, and running. The developed model achieved action-wise accuracy between **80% and 95%**, with overall evaluation metrics of approximately **89% accuracy**, **88% precision**, **87% recall**, and **88% F1-score**. Compared with CNN-only systems, the hybrid model offers stronger sequential understanding and more stable real-time predictions. Therefore, the proposed system is more suitable for surveillance, smart monitoring, and gait-based human activity recognition applications.

| Method | Accuracy (%) | Remarks |
|---|---|---|
| Traditional Gait Recognition | 70 – 78 | Low accuracy, sensitive to noise |
| HMM-Based System | 78 – 82 | Limited feature learning |
| GEI + SVM | 85 – 88 | Depends on handcrafted features |
| Proposed CNN + LSTM | 80 – 95 | Requires training data and higher computation |

**Table 7.1: Comparison with Existing Action/ Gait Recognition Methods**

# CHAPTER 8

# Publication

## 8.1 Publication Details

The project titled **Detection and Recognition of Person Re-Identification through Gait Analysis Using CNN+LSTM** has been successfully published in the **International Journal of Computer Applications (IJCA)**, ISSN **0975–8887**, **Volume 187, Number 25, July 2025**. The publication presents a cross-scene person re-identification system that uses gait recognition as a biometric method for identifying individuals across different camera views. This research was contributed by authors from the **Department of Computer Science and Engineering, Faculty of Engineering and Technology, Jain University, Bengaluru, Karnataka, India**. https://www.ijcaonline.org/archives/volume187/number25/a-cross-scene-person-re-identification-through-gait-recognition-using-silhouette-based-deep-learning-model/

The published work focuses on using deep learning techniques such as **Convolutional Neural Networks (CNN)** and **Recurrent Neural Networks (RNN/LSTM)** to analyze walking patterns from video sequences. The framework extracts human silhouettes, learns spatial gait features through CNN, and captures temporal motion patterns through LSTM networks. The study highlights that gait recognition is a reliable biometric because walking style remains comparatively stable even under changes in clothing, lighting, and camera angles. The system was designed for applications such as surveillance, access control, forensic investigation, and smart monitoring environments.

Experimental evaluation in the publication showed strong performance on benchmark gait datasets, demonstrating high recognition accuracy and robustness in real-world scenarios. The paper concludes that integrating CNN with sequence-learning networks significantly improves person re-identification performance compared to traditional appearance-based methods. It also emphasizes future enhancements such as multi-modal biometrics, lightweight real-time models, and larger real-world datasets. This publication represents an important academic contribution toward non-intrusive and scalable person re-identification systems using gait analysis.

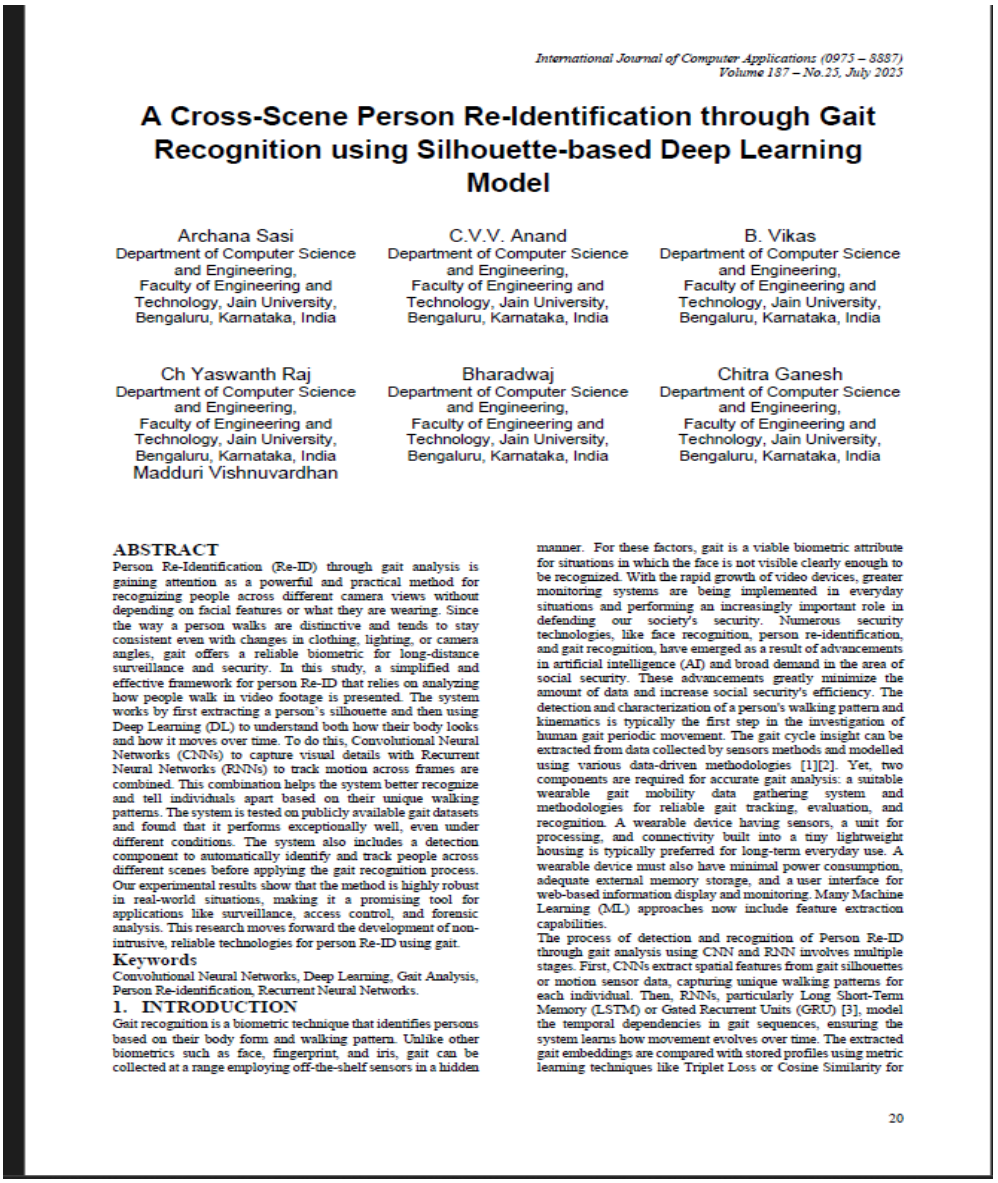
## 8.2 Abstract of the Publication

Person Re-Identification (Re-ID) through gait analysis is gaining attention as a powerful and practical method for recognizing people across different camera views without depending on facial features or what they are wearing. Since the way a person walks are distinctive and tends to stay consistent even with changes in clothing, lighting, or camera angles, gait offers a reliable biometric for long-distance surveillance and security.

In this study, a simplified and effective framework for person Re-ID that relies on analyzing how people walk in video footage is presented. The system works by first extracting a person's silhouette and then using Deep Learning (DL) to understand both how their body looks and how it moves over time. To do this, Convolutional Neural Networks (CNNs) to capture visual details with Recurrent Neural Networks (RNNs) to track motion across frames are combined. This combination helps the system better recognize and tell individuals apart based on their unique walking patterns. The system is tested on publicly available gait datasets and found that it performs exceptionally well, even under different conditions. The system also includes a detection component to automatically identify and track people across different scenes before applying the gait recognition process. Our experimental results show that the method is highly robust in real-world situations, making it a promising tool for applications like surveillance, access control, and forensic analysis. This research moves forward the development of non-intrusive, reliable technologies for person Re-ID using gait.

## 8.3 Publication Proof

The proof of publication is provided below:



*International Journal of Computer Applications (0975 – 8887)*
*Volume 187 – No.25, July 2025*

# A Cross-Scene Person Re-Identification through Gait Recognition using Silhouette-based Deep Learning Model

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**ABSTRACT**
Person Re-Identification (Re-ID) through gait analysis is gaining attention as a powerful and practical method for recognizing people across different camera views without depending on facial features or what they are wearing. Since the way a person walks are distinctive and tends to stay consistent even with changes in clothing, lighting, or camera angles, gait offers a reliable biometric for long-distance surveillance and security. In this study, a simplified and effective framework for person Re-ID that relies on analyzing how people walk in video footage is presented. The system works by first extracting a person's silhouette and then using Deep Learning (DL) to understand both how their body looks and how it moves over time. To do this, Convolutional Neural Networks (CNNs) to capture visual details with Recurrent Neural Networks (RNNs) to track motion across frames are combined. This combination helps the system better recognize and tell individuals apart based on their unique walking patterns. The system is tested on publicly available gait datasets and found that it performs exceptionally well, even under different conditions. The system also includes a detection component to automatically identify and track people across different scenes before applying the gait recognition process. Our experimental results show that the method is highly robust in real-world situations, making it a promising tool for applications like surveillance, access control, and forensic analysis. This research moves forward the development of non-intrusive, reliable technologies for person Re-ID using gait.

**Keywords**
Convolutional Neural Networks, Deep Learning, Gait Analysis, Person Re-identification, Recurrent Neural Networks.

**1. INTRODUCTION**
Gait recognition is a biometric technique that identifies persons based on their body form and walking pattern. Unlike other biometrics such as face, fingerprint, and iris, gait can be collected at a range employing off-the-shelf sensors in a hidden manner. For these factors, gait is a viable biometric attribute for situations in which the face is not visible clearly enough to be recognized. With the rapid growth of video devices, greater monitoring systems are being implemented in everyday situations and performing an increasingly important role in defending our society's security. Numerous security technologies, like face recognition, person re-identification, and gait recognition, have emerged as a result of advancements in artificial intelligence (AI) and broad demand in the area of social security. These advancements greatly minimize the amount of data and increase social security's efficiency. The detection and characterization of a person's walking pattern and kinematics is typically the first step in the investigation of human gait periodic movement. The gait cycle insight can be extracted from data collected by sensors methods and modelled using various data-driven methodologies [1][2]. Yet, two components are required for accurate gait analysis: a suitable wearable gait mobility data gathering system and methodologies for reliable gait tracking, evaluation, and recognition. A wearable device having sensors, a unit for processing, and connectivity built into a tiny lightweight housing is typically preferred for long-term everyday use. A wearable device must also have minimal power consumption, adequate external memory storage, and a user interface for web-based information display and monitoring. Many Machine Learning (ML) approaches now include feature extraction capabilities.
The process of detection and recognition of Person Re-ID through gait analysis using CNN and RNN involves multiple stages. First, CNNs extract spatial features from gait silhouettes or motion sensor data, capturing unique walking patterns for each individual. Then, RNNs, particularly Long Short-Term Memory (LSTM) or Gated Recurrent Units (GRU) [3], model the temporal dependencies in gait sequences, ensuring the system learns how movement evolves over time. The extracted gait embeddings are compared with stored profiles using metric learning techniques like Triplet Loss or Cosine Similarity for

20

**Figure 8.1: First page of the published Journal**

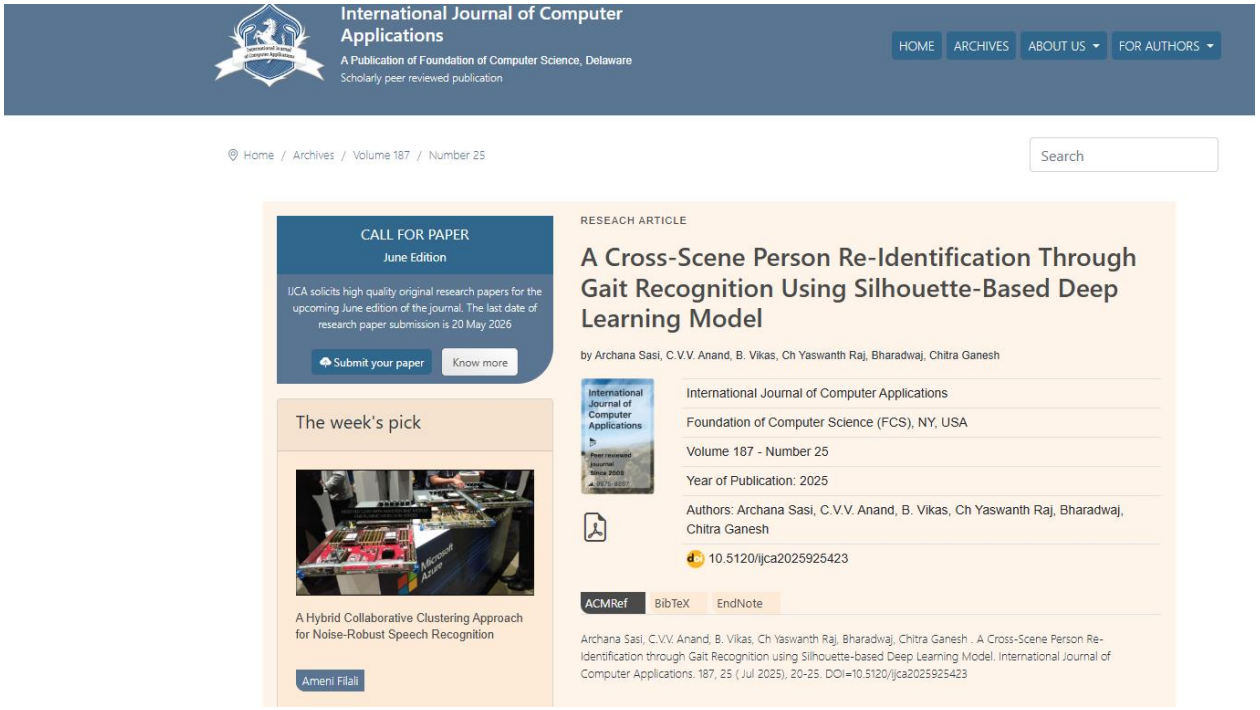


**Figure 8.2: IJCA Online publication page showing journal details**

## 8.4 Significance of Publication

Publishing this project in an international journal provides several important advantages, including academic recognition, increased credibility, and professional value for the research work. It validates the quality of the project through peer review and makes the study accessible to researchers, students, and industry professionals worldwide. A publication also strengthens higher education applications, resumes, job opportunities, and research profiles by demonstrating practical knowledge in artificial intelligence, deep learning, and computer vision. In addition, it helps the authors contribute to technological advancement in gait-based person re-identification and creates opportunities for future collaborations, citations, and further innovations in surveillance and smart security systems.

## 8.5 Journal Submission Status

Along with the open-access publication, the research paper has already been successfully published in the International Journal of Computer Applications (IJCA) after formal review and acceptance. The paper is available in Volume 187, Number 25, July 2025, providing academic recognition and public accessibility. This follows the standard research workflow in which the work undergoes evaluation, publication, and dissemination through a recognized journal platform, ensuring credibility and visibility for future research references.

# CHAPTER 9

## 8.1 CONCLUSION

The project titled Detection and Recognition of Person Re-Identification through Gait Analysis Using CNN + LSTM successfully demonstrates an intelligent approach for identifying individuals and recognizing human actions through walking patterns and motion behavior. By using video input, frame extraction, preprocessing, and deep learning techniques, the system effectively classifies actions such as boxing, handclapping, handwaving, jogging, running, and walking. The combination of gait analysis with action recognition provides a non-intrusive and practical solution for person identification in surveillance environments.

The hybrid CNN + LSTM architecture proved effective in learning both spatial and temporal features from video sequences. CNN extracted body posture, silhouette, and frame-level motion features, while LSTM captured time-based movement patterns across consecutive frames. Experimental results showed strong performance, with action-wise accuracy ranging from 80% to 95%, and evaluation metrics such as accuracy, precision, recall, and F1-score confirming reliable classification performance. The successful deployment through a web platform further proves the system's usability in real-world applications.

In conclusion, the proposed system offers a scalable, efficient, and accurate framework for gait-based person re-identification and human activity recognition. It can be applied in areas such as smart surveillance, security monitoring, forensic analysis, healthcare observation, and access control systems. Future enhancements may include larger datasets, real-time multi-person tracking, mobile optimization, and advanced transformer-based deep learning models to further improve accuracy and speed.

## 8.2 FUTURE SCOPE

The future scope of the project Detection and Recognition of Person Re-Identification through Gait Analysis Using CNN + LSTM is highly promising in the field of intelligent surveillance and biometric security systems. The current system can be further improved by training with larger and more diverse datasets containing different camera angles, lighting conditions, clothing variations, and crowded environments. This would increase the robustness and accuracy of person re-identification in real-world scenarios. Additional action classes and abnormal behavior detection can also be included to expand system capabilities.

Another important future enhancement is the development of real-time multi-person tracking and recognition. The existing model can be integrated with advanced object detection algorithms such as YOLO or OpenCV tracking modules to identify multiple individuals simultaneously across different camera feeds. Deployment on mobile devices, edge devices, and IoT surveillance systems can make the project more practical for smart cities, transportation hubs, offices, and public safety monitoring. Lightweight deep learning models can also be used to reduce computational cost and improve speed.

The future scope also includes integrating modern deep learning architectures such as Transformers, attention mechanisms, and multi-modal biometrics combining gait, face, and posture recognition. This would improve identification reliability even under occlusion or partial visibility conditions. The system may also be applied in healthcare for patient movement monitoring, sports performance analysis, elderly fall detection, and rehabilitation support. With continuous advancements in artificial intelligence, this project has strong potential to evolve into a complete next-generation smart monitoring solution.

# CHAPTER 10

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